

Klann's Leg mechanism

Klann's Linkage

6-bar walking mechanism that simulates the gait of legged animals, invented by Joseph C. Klann (US patents 6260862, 6478314). Just 2 Klann legs can provide a smooth walking motion whereas 3 legs required for the Jansen and Ghassaei linkages. The Klann Linkage is classified as a modified Stephenson-III kinematic chain. Proportions are defined to optimize the linearity of the foot for one half rotation of the crank.

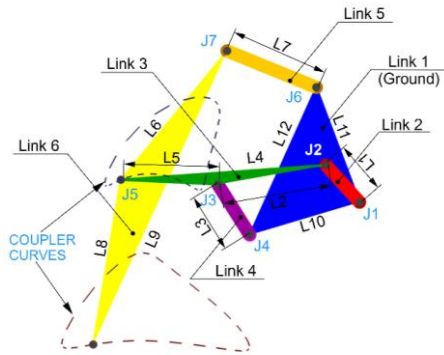


Figure 1: Klann's linkage in MechAnalyzer showing its coupler curve

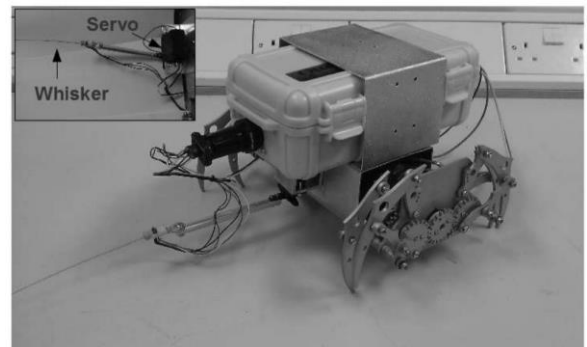


Figure 2: An underwater walking robot (the "machalobster") that uses the Klann's linkage

Walking Mechanisms

- Biped robots present certain challenges due to the large number of actuators: complexities of design & control, low energy efficiency, overly high price complicating practical use.
- A different way to think about biped robots is to use reduced degrees of freedom by way of using a leg mechanism, which can simulate the walking gait of humans and animals.
- Chebyshev was one of the first to suggest a walking machine based on the fourbar linkage: *the Pantigrade Machine*. (Original models available at the Museum of the History of Physics and Mathematics of Saint Petersburg State University, Peterhof, Russia).
- Chebyshev's machine spawned the development of many different leg mechanisms, using many different standard mechanisms as their bases: *the fourbar linkage with a pantograph, the Watt-II linkage, the Stephenson-III linkage, the Klann linkage, the Jansen linkage etc.*

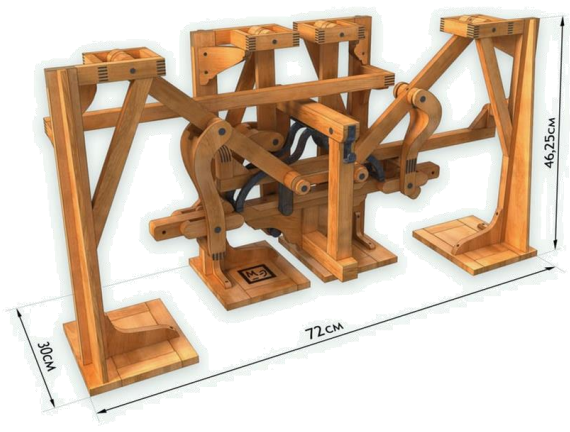


Figure 3: Chebyshev's Pantigrade Walker